

Exercise 10

Advanced SQL 1: Aggregates, joins and recursion

For this exercise sheet, please use the prepared SQL scripts containing the BikeStore data from the lecture DB-Chapter07-KH-Advanced SQL BikeStore Sample DB.zip

Task 1: Joins

In this task, you wonder, if your BikeStore business has some unused potential.

- a) * You wonder, if there are customers that never placed an order and you need their email addresses to send them advertisement.
- b) To get a better reply rate, you decide to let your employees write an email to customers that did place an order with them, but have actually never ordered a product. (So all orders that do not have order items)
- c) * Next thing you wonder, if you need to restock your stores with new items. List all products that have less than 20 items in stock over all stores and that are from 2018 or 2019.

Task 2: Aggregate functions and joins

- a) * Your business with bikes is doing great! But you want to know what the actual turnover per store over all times is, to enjoy how much money you made.
- b) You wonder, if all your employees are working diligently and want to know how many orders per employee there are.
- c) Since that would discriminate employees that have only a few orders, but with large quantities, you want to check if there is a high variance in order prices and the average order prices.
- d) The numbers from the task before doesn't look too good. In order to get a benchmark for the success of an employee, you want to know the average turnover per employee per month.

Task 3: Recursion

Well, some of the employees didn't sell anything. Maybe they are managing people instead of doing sales? You need to know, how many people each person is managing and compare their sales. But this is a complex task and you decide to proceed step by step.

- a) * First, you need a table showing on the left side the name of the person and on the right side who this person is managing. Each tuple is describing one manager-employee relationship. Since Jenette David is managing Layla Terrel indirectly over Venita Daniel, you think about using recursion.
- b) Add a column to the recursive query from before to show whether an employee is directly managed by a person or via another inline manager.
- c) Add for each person managing an employee the next inline manager, so that you know as Fabilola Jackson when you want to manage Layla Terrel, you have to talk to Jannette David.
- d) You also need to know as Fabilola Jackson, that if you have an assignment for Layla Terrel your assignment needs to pass over two managers.
- e) Finally, you can use the recursive query from this task and the last query from the last task to see how many people each employee is managing and what their average turnover per month is. Who is looking suspicious?

Task 4: Cycle detection

For this task, please use the prepared SQL script `DB-Exercise10-KH-Animals.sql` from the repository.

- a) The test script contains the relation `eats`. Create output showing which animal eats what. For example, a monkey eats frogs and a frog eats flies. So a monkey also (indirectly) eats flies. Use recursion to complete the task.
- b) Expand the test data by adding the entry "fly eats wolf". Then recognise this circle in the previously implemented recursion and stop the recursion at the appropriate point.
- c) Expand the recursion so that it is clear whether the animal eats something directly or indirectly. For example, the fly eats sugar water directly, but the monkey only eats flies indirectly by eating frogs.
- d) Expand the recursion so that it is clear which animal needs to eat which other animal in order to eat something specific. For example, the monkey could eat frogs, and then it would have the chance to consume flies as part of its diet without having to eat them directly.